

21-OCT-2024

SUSPECTED UNEXPECTED SERIOUS ADVERSE REACTION (SUSAR) REPORT

SUSAR Report for Nivolumab (BMS-936558), # BMS-2020-107980

Reference Document: Investigator Brochure, Version 19 (29-Jun-2020)

Protocol: CA224-020

Dear Investigator:

As the sponsor of clinical studies with the investigational medicinal product (IMP) specified above, Bristol-Myers Squibb Research & Development is issuing this safety report. The details of this adverse event/safety issue are being reported to you in compliance with regulations.

For Suspected Unexpected Serious Adverse Reactions (SUSARs) occurring in blinded clinical trials, study treatments will remain blinded in reports sent to investigators.

The attached report does not reflect a conclusion or constitute an admission by the sponsor that the IMP caused or contributed to this adverse event/safety issue.

Your Institutional Review Board (IRB)/Ethics Committee (EC) must receive a copy of this SUSAR as per applicable regulation and local IRB/EC policy. Please consult the clinical trial personnel or documents/manuals regarding your responsibility for submitting SUSARs to your IRB/EC. Please also review your IRB/EC procedures relating to adverse events/safety issues that may necessitate changes to the Informed Consent Form.

Please include this SUSAR with all copies of the Reference Document.

We thank you in advance for your prompt attention to this information.

Karthik Muthusami

Head, Expedited Safety Reporting

Worldwide Patient Safety

Attachment: CIOMS Report

SUSPECT ADVERSE REACTION REPORT												
	BMS-2020-107980(8)											

I. REACTION INFORMATION

1. PATIENT INITIALS (first, last) [REDACTED]	1a. COUNTRY FRANCE	2. DATE OF BIRTH			2a. AGE Years 43	3. SEX Male	4-6 REACTION ONSET			8-12 CHECK ALL APPROPRIATE TO ADVERSE REACTION
		Day [REDACTED]	Month [REDACTED]	Year [REDACTED]			Day 16	Month Dec	Year 2020	
7+13 DESCRIBE REACTION(S) (including relevant tests/lab data) 1) Immuno-induced diabetes (English) MedDRA Version : v.27.0 IMMUNO-INDUCED DIABETES (Diabetes mellitus (10012601), Diabetes mellitus (10012601)) (16/Dec/2020 -) - Not Recovered/Not Resolved/Ongoing 2) insulin pump implementation due to diabetes mellitus (English) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS (Insulin therapy (10085221), Insulin therapy (10085221)) (05/Jul/2021 - 13/Jul/2021) - Recovered/Resolved Cont..										
										☐ PATIENT DIED ☐ LIFE THREATENING ☒ INVOLVED OR PROLONGED INPATIENT HOSPITALIZATION ☐ RESULTS IN PERSISTENCE OR SIGNIFICANT DISABILITY/INCAPACITY ☐ CONGENITAL ANOMALY ☒ OTHER MEDICALLY IMPORTANT CONDITION

II. SUSPECT DRUG(S) INFORMATION

14. SUSPECT DRUG(S)(include generic name) 1) RELATLIMAB (RELATLIMAB) (Suspect) (Verum) (Solution for injection)(Unknown) Cont..		20. DID EVENT ABATE AFTER STOPPING DRUG? ☐ YES ☐ NO ☒ NA 21. DID EVENT REAPPEAR AFTER REINTRODUCTION ☐ YES ☐ NO ☒ NA (NA : Not Applicable)
15. DAILY DOSE(S) 1) 5.714 milligram(s) (160 milligram(s), 1 in 4 Week)	16. ROUTE(S) OF ADMINISTRATION 1) Intravenous (not otherwise specified)	
17. INDICATION(S) FOR USE 1) MELANOMA [10053571 - Melanoma] Cont..		
18. THERAPY DATE(S) (from/to) 1) (03/Dec/2019 - Unknown)	19. THERAPY DURATION Unknown	

III. CONCOMITANT DRUG(S) AND HISTORY

22. CONCOMITANT DRUG(S) AND DATES OF ADMINISTRATION (exclude those used to treat reaction) 1) ELIQUIS(APIXABAN)(16/May/2020 - Unknown) Cont..
23. OTHER RELEVANT HISTORY (e.g. diagnostics, allergies, pregnancy with last month of period, etc.) MedDRA Version : v.27.0 1) LEFT KNEE LIGAMENOTPLASTY (10023475, Knee ligament repair) (//2000 - Unknown) (Continuing: No) Cont..

IV. MANUFACTURER INFORMATION

24a. NAME AND ADDRESS OF MANUFACTURER Name : BMS UNITED STATES OF AMERICA aepbusinessprocess@bms.com		Study Information Study Name: A Phase 1/2a Dose Escalation and C (Cont..) EudraCT Number: 2014-002605-38 Protocol No.: CA224-020 Center No.: 0038 Subject Id : 01876	
24. REPORT NULLIFIED ☐ YES ☐ NO	24b. MFR CONTROL NO. BMS-2020-107980(8)		
24c. DATE RECEIVED BY MANUFACTURER 12/Oct/2024	24d. REPORT SOURCE ☒ STUDY ☐ LITERATURE ☒ HEALTH PROFESSIONAL		
DATE OF THIS REPORT 18/Oct/2024	25a. REPORT TYPE ☐ INITIAL ☒ FOLLOWUP		

= Continuation attached sheet(s)..

Continuation Sheet for CIOMS report

7+13 DESCRIBE REACTION(S) (including relevant tests/lab data) (Continuation...)

Event Description :

Protocol title (CA224-020): A Phase 1/2a Dose Escalation and Cohort Expansion Study of the Safety, Tolerability, and Efficacy of Anti-LAG-3 Monoclonal Antibody (BMS-986016) Administered Alone and in Combination with Anti-PD-1 Monoclonal Antibody (Nivolumab, BMS-936558) in Advanced Solid Tumors

PID: CA224-020-0038-01876 / BMS-2020-107980 / France

Reportable SAEs: Diabetes Mellitus (Verbatim: Immuno-Induced Diabetes) - SUSAR for relatlimab; and Insulin Therapy (Verbatim: Insulin Pump Implementation due to Diabetes Mellitus) - SUSAR for relatlimab and nivolumab

A clinical investigator reported that a 43-year-old male patient was hospitalized due to CTC grade 3 diabetes mellitus, 15 days after the most recent dose of nivolumab and relatlimab study therapy for melanoma. Approximately six months later, the patient was re-hospitalized for insulin pump implementation (CTC grade 1). The patient received treatment for diabetes mellitus. At the time of this report, the event diabetes mellitus was ongoing. Insulin pump implementation was completed. Study therapy dose was delayed due to the event diabetes mellitus whereas no action was taken due to insulin pump implementation.

The patient's medical history included ligamentoplasty in 2000 (left knee) and deep venous thrombosis (left lower limb). He had no family or medical history of diabetes mellitus.

The patient's concomitant medication included apixaban.

On 26-Nov-2019, the patient's baseline laboratory test results showed haemoglobin A1c (HbA1c) level at 5.7% (reference range [RR]: 4-6 %) and glucose level 4.29 mmol/L (RR: 4.11-5.89 mmol/L).

On 30-Nov-2019, his glucose level was 5.20 mmol/L (RR: 4.10-6.05 mmol/L).

On 03-Dec-2019, open-label study therapy was initiated with intravenous (IV) nivolumab (480 mg; every 4 weeks [Q4WK]) and IV relatlimab (160 mg; Q4WK). Prior to the onset of the events, the patient received his most recent dose of study therapy on 01-Dec-2020.

From Nov-2019 to Sep-2020, his glucose levels were within the upper limit normal of RR.

On 07-Sep-2020, his glucose level was 0.98 g/L (RR: 0.74-1.09 g/L).

On 05-Oct-2020, the patient's glucose level was 1.14 g/L.

On 02-Nov-2020, his glucose level was 1.14 g/L.

On 30-Nov-2020, the patient's laboratory test results showed elevated level of blood glucose at 1.94 g/L (RR: 0.74-0.9 g/L).

Since 01-Dec-2020, his fasting blood glucose levels were elevated (values were not provided).

On 16-Dec-2020, the patient was hospitalized due to immuno-induced diabetes. Treatment with insulin lispro and insulin glargine was initiated.

On 17-Dec-2020, his laboratory test results showed HbA1c level at 7.7%, C-peptide level 0.324 nmol/L (RR: 0.370-1.47 nmol/L), glutamate decarboxylase (GAD65A) was less than 0.70 U/mL (negative), anti-islet cell antibody (ICA) was less than 0.80 U/mL (negative) and zinc transporter 8 autoantibodies (ZnT8Ab) was 5.9 UA/ML (negative) (RR: less than 15 UA/ML). No computed tomography (CT)/magnetic resonance imaging (MRI) scan was performed during this hospitalization. CT scan was planned on 28-Dec-2020 (as per protocol).

On 21-Dec-2020, treatment with insulin glargine was completed and was started on insulin degludec injection (14 UI).

On 22-Dec-2020, insulin lispro was administered as 2.5 UI at 0800 AM, 3 UI at 1200 PM and 5 UI at 0600 PM.

From 05-Jul-2021 through 13-Jul-2021, the patient had a planned re-hospitalization at the endocrinology department for the installation of an insulin pump and therapeutic education. It was confirmed that during this hospitalization, the immuno-induced diabetes was still ongoing and there was no worsening.

On 13-Jul-2021, insulin therapy was completed.

From 05-Dec-2021 till 11-Dec-2021, the patient was re-hospitalized for therapeutic education regarding functional insulin therapy.

According to the endocrinologist, diagnosis was immune-induced diabetes, which was well controlled after hospitalization from 16-Dec-2020 to 22-Dec-2020.

According to the investigator, diagnosis was confirmed as immuno-induced diabetes, insulin dependent diabetes but not type 1 or type 2. Immuno-induced diabetes was considered related to study therapy since before the start of study therapy, blood glucose levels were normal, which then

Continuation Sheet for CIOMS report

gradually increased with the start of study therapy. The patient did not have pancreatic abnormalities/metastasis, had no recent change in concomitant medications and he did not receive any steroids or herbal supplements.

Supplemental information received on 12-Oct-2024 from the clinical investigator reported that insulin therapy was completed on 13-Jul-2021, which has been incorporated into the above narrative.

Investigator causality assessment: Immuno-induced diabetes and insulin therapy were related to nivolumab and relatlimab study therapy.

BMS causality assessment: Immuno-induced diabetes was related, and insulin therapy was not related to nivolumab and relatlimab study therapy.

Expectedness Assessment:

Nivolumab (IB): Diabetes mellitus was listed, and insulin therapy was considered unlisted.

Relatlimab (IB): Diabetes mellitus and insulin therapy were considered unlisted.

Company Remarks (Sender's Comments) :

This middle-aged patient had elevated blood glucose levels, reported as immuno-induced diabetes approximately one year after the initiation and two weeks after the most recent dose of nivolumab and relatlimab study therapy for melanoma. His baseline glucose/HbA1C values were normal and he had no family/medical history of diabetes. During this hospitalization, C-peptide level was low, elevated HbA1C and blood glucose values were noted along with negative GAD and islet cell antibodies. Post insulin treatment, the glucose levels were well controlled per endocrinologist. Investigator excluded multiple alternative etiologies and confirmed the diagnosis as immuno-induced diabetes. Approximately six months after the diagnosis of diabetes, patient was re-hospitalized for insulin pump implementation for immuno-induced diabetes. Based on this information with exclusion of multiple alternative etiologies and considering the blood glucose/HbA1C levels showed an upward trend post initiation of study therapy and as type1 diabetes mellitus has been associated with relatlimab and nivolumab therapy, the reported event is considered related to study therapy due to their immune mediated mechanism of action. Insulin pump implementation was a treatment procedure for the event of diabetes, hence, considered unrelated to study therapy.

Additional Information (Continuation...)

Laboratory Data :

See Event Description for additional lab results

Lab Result :

Test Name	Test Date	Test Result	Normal Value
ANTI-ISLET CELL ANTIBODY	17/Dec/2020		
BLOOD GLUCOSE	26/Nov/2019	4.29 millimole per litre	4.11 - 5.89
BLOOD GLUCOSE	30/Nov/2019	5.20 millimole per litre	4.10 - 6.05
BLOOD GLUCOSE	07/Sep/2020	0.98 gram per litre	0.74 - 1.09
BLOOD GLUCOSE	05/Oct/2020	1.14 gram per litre	0.74 - 1.09
BLOOD GLUCOSE	02/Nov/2020	1.14 gram per litre	0.74 - 1.09
BLOOD GLUCOSE	30/Nov/2020	1.94 gram per litre	0.74 - 0.9
C-PEPTIDLE	17/Dec/2020	0.324 nanomole per litre	0.370 - 1.47
FASTING GLYCEMIA	/Nov/2019		
FASTING GLYCEMIA	/Sep/2020		
FASTING GLYCEMIA	01/Dec/2020		

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GAD65A	17/Dec/2020		
HBA1C	26/Nov/2019	5.7 % percent	4 - 6
HBA1C	17/Dec/2020	7.7 % percent	4 - 6
ZINC TRANSPORTER 8 AUTOANTIBODIES	17/Dec/2020		

Test Result (Code) / Result Unstructured Data (free text) :

1) Test Name: ANTI-ISLET CELL ANTIBODY

Result Unstructured Data (free text) : Lesser than 0.80 U/mL or Negative

Test Date: 17/Dec/2020

9) Test Name: FASTING GLYCEMIA

Result Unstructured Data (free text) : From Nov-2019 to Sep-2020, his glucose levels were within the upper limit normal of RR.

Test Date: /Nov/2019

10) Test Name: FASTING GLYCEMIA

Result Unstructured Data (free text) : were within the upper limit normal of RR

Test Date: /Sep/2020

11) Test Name: FASTING GLYCEMIA

Result Unstructured Data (free text) : were elevated (values not provided)

Test Date: 01/Dec/2020

12) Test Name: GAD65A

Result Unstructured Data (free text) : less than 0.70 U/mL or Negative

Test Date: 17/Dec/2020

15) Test Name: ZINC TRANSPORTER 8 AUTOANTIBODIES

Result Unstructured Data (free text) : 5.9 UA/mL (Reference range: less than 15 UA/ML)

Test Date: 17/Dec/2020

14.SUSPECT DRUG(S) (Continuation...)

Product-Reaction Level

1) Drug : RELATLIMAB
 Active Substance : 1) RELATLIMAB
 Coding Class : Verum
 Drug Characterization : Suspect
 Form of Admin : 1) Solution for injection
 Lot Number : 1) Unknown
 Daily Dose : 1) 5.714 milligram(s) (160 milligram(s), 1 in 4 Week)
 Route of Admin : 1) Intravenous (not otherwise specified)
 Indications : 1) MELANOMA [10053571 - Melanoma]
 Therapy Dates : 1) From : 03/Dec/2019 To :Unknown
 Action(s) Taken With Drug : Drug Interrupted

Causality

1) IMMUNO-INDUCED DIABETES (Diabetes mellitus - 10012601, Diabetes mellitus - 10012601)

Causality as per reporter : Related

Causality as per Mfr : Related

DeChallenge : Not applicable

ReChallenge : Not Applicable

2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS (Insulin therapy - 10085221, Insulin therapy - 10085221)

Causality as per reporter : Related

Causality as per Mfr : Not Related

DeChallenge : Not applicable

ReChallenge : Not Applicable

Labeling :

1) IMMUNO-INDUCED DIABETES

IB UnLabeled

2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS

Continuation Sheet for CIOMS report

IB UnLabeled

2) Drug : NIVOLUMAB
 Active Substance : 1) NIVOLUMAB
 Coding Class : Verum
 Drug Characterization : Suspect
 Form of Admin : 1) Solution for injection
 Lot Number : 1) Unknown
 Daily Dose : 1) 17.14 milligram(s) (480 milligram(s), 1 in 4 Week)
 Route of Admin : 1) Intravenous (not otherwise specified)
 Indications : 1) Melanoma [10053571 - Melanoma]
 Therapy Dates : 1) From : 03/Dec/2019 To :Unknown
 Action(s) Taken With Drug : Drug Interrupted

Causality

1) IMMUNO-INDUCED DIABETES (Diabetes mellitus - 10012601, Diabetes mellitus - 10012601)
 Causality as per reporter : Related
 Causality as per Mfr : Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable
 2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS (Insulin therapy - 10085221, Insulin therapy - 10085221)
 Causality as per reporter : Related
 Causality as per Mfr : Not Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable

Labeling :

1) IMMUNO-INDUCED DIABETES
 CORE Labeled
 IB Labeled
 UNITED STATES OF AMERICA Labeled
 2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS
 CORE Labeled
 IB UnLabeled
 UNITED STATES OF AMERICA Labeled

3) Drug : RELATLIMAB_JPN
 Active Substance : 1) RELATLIMAB
 Drug Characterization : Drug Not Administered
 Form of Admin : 1) Injection
 Lot Number : 1) Unknown
 Daily Dose : 1) 5.714 milligram(s) (160 milligram(s), 1 in 4 Week)
 Route of Admin : 1) Intravenous (not otherwise specified)
 Indications : 1) melanoma [10053571 - Melanoma]
 Therapy Dates :
 Action(s) Taken With Drug : Not applicable

Causality

1) IMMUNO-INDUCED DIABETES (Diabetes mellitus - 10012601, Diabetes mellitus - 10012601)
 Causality as per reporter : Related
 Causality as per Mfr : Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable
 2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS (Insulin therapy - 10085221, Insulin therapy - 10085221)
 Causality as per reporter : Related
 Causality as per Mfr : Not Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable

Labeling :

1) IMMUNO-INDUCED DIABETES
 IB UnLabeled
 2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS
 IB UnLabeled
 4) Drug : RELATLIMAB,NIVOLUMAB_JPN
 Active Substance : 1) RELATLIMAB

Continuation Sheet for CIOMS report

2) NIVOLUMAB
 Drug Characterization : Drug Not Administered
 Form of Admin : 1) Injection
 Lot Number : 1) Unknown
 Daily Dose : 1) 5.714 milligram(s) (160 milligram(s), 1 in 4 Week)
 Route of Admin : 1) Intravenous (not otherwise specified)
 Indications : 1) melanoma [10053571 - Melanoma]
 Therapy Dates :
 Action(s) Taken With Drug : Not applicable

Causality

- 1) IMMUNO-INDUCED DIABETES (Diabetes mellitus - 10012601, Diabetes mellitus - 10012601)
 Causality as per reporter : Related
 Causality as per Mfr : Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable
- 2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS (Insulin therapy - 10085221, Insulin therapy - 10085221)
 Causality as per reporter : Related
 Causality as per Mfr : Not Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable

Labeling :

- 1) IMMUNO-INDUCED DIABETES
 IB UnLabeled
- 2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS
 IB UnLabeled

- 5) Drug : RELATLIMAB,NIVOLUMAB_JPN
 Active Substance : 1) RELATLIMAB
 2) NIVOLUMAB
 Drug Characterization : Drug Not Administered
 Form of Admin : 1) Injection
 Lot Number : 1) Unknown
 Daily Dose : 1) 17.14 milligram(s) (480 milligram(s), 1 in 4 Week)
 Route of Admin : 1) Intravenous (not otherwise specified)
 Indications : 1) melanoma [10053571 - Melanoma]
 Therapy Dates :
 Action(s) Taken With Drug : Not applicable

Causality

- 1) IMMUNO-INDUCED DIABETES (Diabetes mellitus - 10012601, Diabetes mellitus - 10012601)
 Causality as per reporter : Related
 Causality as per Mfr : Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable
- 2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS (Insulin therapy - 10085221, Insulin therapy - 10085221)
 Causality as per reporter : Related
 Causality as per Mfr : Not Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable

Labeling :

- 1) IMMUNO-INDUCED DIABETES
 IB UnLabeled
- 2) INSULIN PUMP IMPLEMENTATION DUE TO DIABETES MELLITUS
 IB UnLabeled

15. DAILY DOSE(S) (Continuation...)

Dosage Text :

Drug 1 :RELATLIMAB

1) Prior to the onset of the events, the patient received his most recent dose of study therapy on 01-Dec-2020.

Drug 2 :NIVOLUMAB

1) Prior to the onset of the events, the patient received his most recent dose of study therapy on 01-Dec-2020.

Continuation Sheet for CIOMS report

22.CONCOMITANT DRUG(S) (Continuation...)

1). Drug : ELIQUIS
 Active Substance : 1) APIXABAN
 Form Strength :
 Form of Admin : 1) Unknown
 Route of Admin : 1) Unknown
 Indications : 1) PRODUCT USED FOR UNKNOWN INDICATION [10070592 - Product used for unknown indication]
 Therapy Dates : 1) From : 16/May/2020 To : Unknown

23. OTHER RELEVANT HISTORY (Continuation...)

2) LEFT LOWER LIMB DEEP VENOUS THROMBOSIS (10043630 , Thrombosis of leg deep venous) (22/Nov/2019 - Unknown) (Continuing : Unknown)

3) INSULIN PUMP INSTALLATION (10085222 , Insulin pump therapy) (05/Jul/2021 - 13/Jul/2021) (Continuing : NO)

4) MELANOMA (10053571 , Melanoma) (Continuing : YES)

Study Information (Continuation...)

Study Name :

A Phase 1/2a Dose Escalation and Cohort Expansion Study of the Safety, Tolerability, and Efficacy of Anti-LAG-3 Monoclonal Antibody (BMS-986016) Administered Alone and in Combination with Anti-PD-1 Monoclonal Antibody (Nivolumab, BMS-936558) in Advanced Solid Tumors

24. Other Study Registration Numbers

Registration Number	Registration Country
2023-508067-70-00	European Union
NL57142.031.16	NETHERLANDS
2266	GERMANY

Primary Reporter:

